American Computer Science League

2022-2023 ● Contest 3: Short Problems Solutions ● Junior Division

1. Boolean Algebra

$$(A + B) + AB + B = A + B + AB$$
$$= A + B(1 + A)$$
$$= A + B$$

D. $A + B$

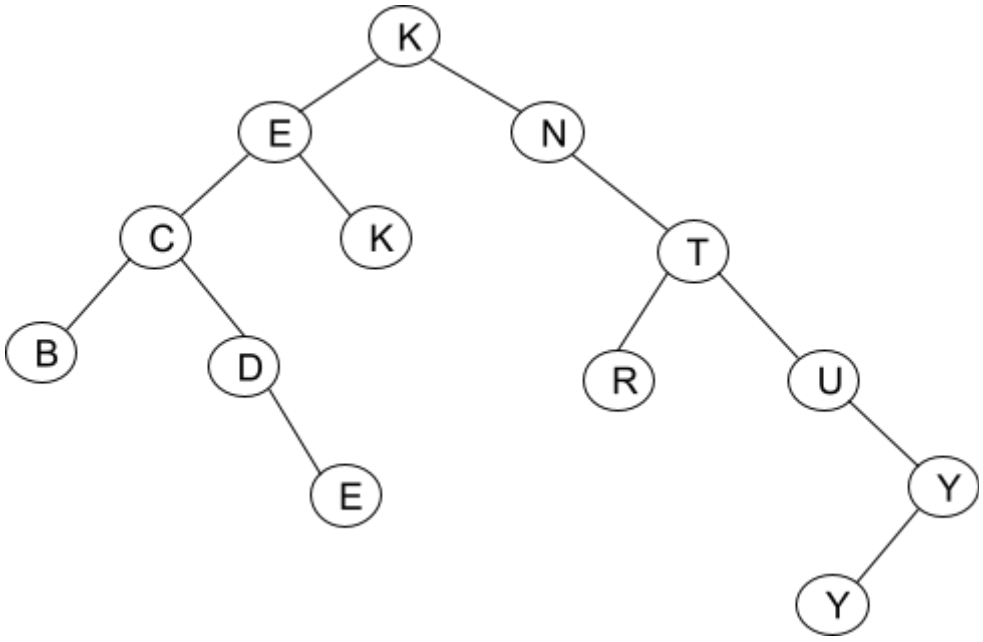
2. Boolean Algebra

$$(\overline{A}\,B)(A + \overline{B}) = \overline{A}\,BA + \overline{A}\,B\,\overline{B} = 0\,\overline{A}\,B + 0 = 0$$

A. 0

3. Data Structures

The binary search tree for K E N T U C K Y D E R B Y is:



The nodes at depth 3 from left to right are BDRU.

B. BDRU

4. Data Structures The stack is constructed using LIFO as follows: M, ME, MER, ME, M, MC, MCH, MC, M, MA, M, MN, MNT, MNTO, MNTOF, MNTO, MNTOV, MNTO, MNTOE, MNTOEN, MNTOE, MNTO, MNTOI, MNTO, MNTOC, MNTOCE, MNTOCE, MNTOC, MNTO The next item popped would be: O	D. O
5. What Does This Program Do? (Arrays) The sum of the elements of val is 45. This program calculates the standard deviation of the array. a=5, t=1^2+2^2+3^2+1^2+2^2+2^2+3^2+2^2+0^2=36, sqrt(36/ 9)=2	A. 2